

Appendix B *Axioms and Proof Rules*

The following axioms and proof rules were used in the proof systems studied in this book.

AXIOM 1: SKIP

$$\{p\} \text{ skip } \{p\}$$

AXIOM 2: ASSIGNMENT

$$\{p[u := t]\} \ u := t \ \{p\}$$

where $u \in Var$ or $u \equiv a[s_1, \dots, s_n]$, for $a \in Var$.

AXIOM 2': PARALLEL ASSIGNMENT

$$\{p[\bar{x} := \bar{t}]\} \ \bar{x} := \bar{t} \ \{p\}$$

RULE 3: COMPOSITION

$$\frac{\{p\} \ S_1 \ \{r\}, \{r\} \ S_2 \ \{q\}}{\{p\} \ S_1; \ S_2 \ \{q\}}$$

RULE 4: CONDITIONAL

$$\frac{\{p \wedge B\} \ S_1 \ \{q\}, \{p \wedge \neg B\} \ S_2 \ \{q\}}{\{p\} \ \text{if } B \text{ then } S_1 \ \text{else } S_2 \ \text{fi } \{q\}}$$

RULE 4': FAILURE

$$\frac{\{p \wedge B\} \ S \ \{q\}}{\{p\} \ \text{if } B \rightarrow S \ \text{fi } \{q\}}$$

RULE 4'': FAILURE II

$$\frac{p \rightarrow B, \{p\} S \{q\}}{\{p\} \text{ if } B \rightarrow S \text{ fi } \{q\}}$$

RULE 5: LOOP

$$\frac{\{p \wedge B\} S \{p\}}{\{p\} \text{ while } B \text{ do } S \text{ od } \{p \wedge \neg B\}}$$

RULE 6: CONSEQUENCE

$$\frac{p \rightarrow p_1, \{p_1\} S \{q_1\}, q_1 \rightarrow q}{\{p\} S \{q\}}$$

RULE 7: LOOP II

$$\frac{\begin{array}{l} \{p \wedge B\} S \{p\}, \\ \{p \wedge B \wedge t = z\} S \{t < z\}, \\ p \rightarrow t \geq 0 \end{array}}{\{p\} \text{ while } B \text{ do } S \text{ od } \{p \wedge \neg B\}}$$

where t is an integer expression and z is an integer variable that does not appear in p, B, t or S .

RULE 8: RECURSION

$$\frac{\begin{array}{l} \{p_1\} P_1 \{q_1\}, \dots, \{p_n\} P_n \{q_n\} \vdash \{p\} S \{q\}, \\ \{p_1\} P_1 \{q_1\}, \dots, \{p_n\} P_n \{q_n\} \vdash \{p_i\} S_i \{q_i\}, i \in \{1, \dots, n\} \end{array}}{\{p\} S \{q\}}$$

where $D = P_1 :: S_1, \dots, P_n :: S_n$.

RULE 9: RECURSION II

$$\frac{\begin{array}{l} \{p_1\} P_1 \{q_1\}, \dots, \{p_n\} P_n \{q_n\} \vdash \{p\} S \{q\}, \\ \{p_1 \wedge t < z\} P_1 \{q_1\}, \dots, \{p_n \wedge t < z\} P_n \{q_n\} \vdash \\ \{p_i \wedge t = z\} S_i \{q_i\}, i \in \{1, \dots, n\}, \\ p_i \rightarrow t \geq 0, i \in \{1, \dots, n\} \end{array}}{\{p\} S \{q\}}$$

where $D = P_1 :: S_1, \dots, P_n :: S_n$ and z is an integer variable that does not occur in p_i, t, q_i and S_i for $i \in \{1, \dots, n\}$ and is treated in the proofs as a constant.

RULE 10: BLOCK

$$\frac{\{p\} \bar{x} := \bar{t}; S \{q\}}{\{p\} \textbf{begin local } \bar{x} := \bar{t}; S \textbf{end } \{q\}}$$

where $\{\bar{x}\} \cap \text{free}(q) = \emptyset$.

RULE 11: INSTANTIATION

$$\frac{\{p\} P(\bar{x}) \{q\}}{\{p[\bar{x} := \bar{t}]\} P(\bar{t}) \{q[\bar{x} := \bar{t}]\}}$$

where $\text{var}(\bar{x}) \cap \text{var}(D) = \text{var}(\bar{t}) \cap \text{change}(D) = \emptyset$.

RULE 12: RECURSION III

$$\frac{\begin{array}{l} \{p_1\} P_1(\bar{t}_1) \{q_1\}, \dots, \{p_n\} P_n(\bar{t}_n) \{q_n\} \vdash \{p\} S \{q\}, \\ \{p_1\} P_1(\bar{t}_1) \{q_1\}, \dots, \{p_n\} P_n(\bar{t}_n) \{q_n\} \vdash \\ \{p_i\} \textbf{begin local } \bar{u}_i := \bar{t}_i; S_i \textbf{end } \{q_i\}, \quad i \in \{1, \dots, n\} \end{array}}{\{p\} S \{q\}}$$

where $P_i(\bar{u}_i) :: S_i \in D$ for $i \in \{1, \dots, n\}$.

RULE 12': MODULARITY

$$\frac{\begin{array}{l} \{p_0\} P(\bar{x}) \{q_0\} \vdash \{p_0\} \textbf{begin local } \bar{u} := \bar{x}; S \textbf{end } \{q_0\}, \\ \{p_0\} P(\bar{x}) \{q_0\}, \{p\} P(\bar{x}) \{q\} \vdash \{p\} \textbf{begin local } \bar{u} := \bar{x}; S \textbf{end } \{q\} \end{array}}{\{p\} P(\bar{x}) \{q\}}$$

where $\text{var}(\bar{x}) \cap \text{var}(D) = \emptyset$ and $D = P(\bar{u}) :: S$.

RULE 13: RECURSION IV

$$\frac{\begin{array}{l} \{p_1\} P_1(\bar{e}_1) \{q_1\}, \dots, \{p_n\} P_n(\bar{e}_n) \{q_n\} \vdash_{tot} \{p\} S \{q\}, \\ \{p_1 \wedge t < z\} P_1(\bar{e}_1) \{q_1\}, \dots, \{p_n \wedge t < z\} P_n(\bar{e}_n) \{q_n\} \vdash_{tot} \\ \{p_i \wedge t = z\} \textbf{begin local } \bar{u}_i := \bar{e}_i; S_i \textbf{end } \{q_i\}, \quad i \in \{1, \dots, n\} \end{array}}{\{p\} S \{q\}}$$

where $P_i(\bar{u}_i) :: S_i \in D$, for $i \in \{1, \dots, n\}$, and z is an integer variable that does not occur in p_i, t, q_i and S_i for $i \in \{1, \dots, n\}$ and is treated in the proofs as a constant.

AXIOM 14: ASSIGNMENT TO INSTANCE VARIABLES

$$\{p[u := t]\} u := t \{p\}$$

where $u \in IVar$ or $u \equiv a[s_1, \dots, s_n]$ and $a \in IVar$.

RULE 15: INSTANTIATION II

$$\frac{\{p\} \ y.m \ \{q\}}{\{p[y := s]\} \ s.m \ \{q[y := s]\}}$$

where $y \notin \text{var}(D)$ and $\text{var}(s) \cap \text{change}(D) = \emptyset$.

RULE 16: RECURSION V

$$\frac{\begin{array}{l} \{p_1\} \ s_1.m_1 \ \{q_1\}, \dots, \{p_n\} \ s_n.m_n \ \{q_n\} \vdash \{p\} \ S \ \{q\}, \\ \{p_1\} \ s_1.m_1 \ \{q_1\}, \dots, \{p_n\} \ s_n.m_n \ \{q_n\} \vdash \\ \quad \{p_i\} \ \mathbf{begin \ local \ this} := s_i; S_i \ \mathbf{end} \ \{q_i\}, \ i \in \{1, \dots, n\} \end{array}}{\{p\} \ S \ \{q\}}$$

where $m_i :: S_i \in D$ for $i \in \{1, \dots, n\}$.

RULE 17: RECURSION VI

$$\frac{\begin{array}{l} \{p_1\} \ s_1.m_1 \ \{q_1\}, \dots, \{p_n\} \ s_n.m_n \ \{q_n\} \vdash \{p\} \ S \ \{q\}, \\ \{p_1 \wedge t < z\} \ s_1.m_1 \ \{q_1\}, \dots, \{p_n \wedge t < z\} \ s_n.m_n \ \{q_n\} \vdash \\ \quad \{p_i \wedge t = z\} \ \mathbf{begin \ local \ this} := s_i; S_i \ \mathbf{end} \ \{q_i\}, \ i \in \{1, \dots, n\} \end{array}}{\{p\} \ S \ \{q\}}$$

where $m_i :: S_i \in D$, for $i \in \{1, \dots, n\}$, and z is an integer variable that does not occur in p_i, t, q_i and S_i for $i \in \{1, \dots, n\}$ and is treated in the proofs as a constant.

RULE 18: INSTANTIATION III

$$\frac{\{p\} \ y.m(\bar{x}) \ \{q\}}{\{p[y, \bar{x} := s, \bar{t}]\} \ s.m(\bar{t}) \ \{q[y, \bar{x} := s, \bar{t}]\}}$$

where $y, \bar{x}$ is a sequence of simple variables in Var which do not appear in D and $\text{var}(s, \bar{t}) \cap \text{change}(D) = \emptyset$.

RULE 19: RECURSION VII

$$\frac{\begin{array}{l} \{p_1\} \ s_1.m_1(\bar{t}_1) \ \{q_1\}, \dots, \{p_n\} \ s_n.m_n(\bar{t}_n) \ \{q_n\} \vdash \{p\} \ S \ \{q\}, \\ \{p_1\} \ s_1.m_1(\bar{t}_1) \ \{q_1\}, \dots, \{p_n\} \ s_n.m_n(\bar{t}_n) \ \{q_n\} \vdash \\ \quad \{p_i\} \ \mathbf{begin \ local \ this}, \bar{u}_i := s_i, \bar{t}_i; S_i \ \mathbf{end} \ \{q_i\}, \ i \in \{1, \dots, n\} \end{array}}{\{p\} \ S \ \{q\}}$$

where $m_i(\bar{u}_i) :: S_i \in D$ for $i \in \{1, \dots, n\}$.

RULE 20: RECURSION VIII

$$\frac{
\begin{array}{l}
\{p_1\} s_1.m_1(\bar{e}_1) \{q_1\}, \dots, \{p_n\} s_n.m_n(\bar{e}_n) \{q_n\} \vdash \{p\} S \{q\}, \\
\{p_1 \wedge t < z\} s_1.m_1(\bar{e}_1) \{q_1\}, \dots, \{p_n \wedge t < z\} s_n.m_n(\bar{e}_n) \{q_n\} \vdash \\
\quad \{p_i \wedge t = z\} \textbf{begin local this}, \bar{u}_i := s_i, \bar{e}_i; S_i \textbf{end} \{q_i\}, i \in \{1, \dots, n\} \\
p_i \rightarrow s_i \neq \textbf{null}, i \in \{1, \dots, n\}
\end{array}
}{
\{p\} S \{q\}
}$$

where $m_i(\bar{u}_i) :: S_i \in D$, for $i \in \{1, \dots, n\}$, and z is an integer variable that does not occur in p_i, t, q_i and S_i for $i \in \{1, \dots, n\}$ and is treated in the proofs as a constant.

AXIOM 21: OBJECT CREATION

$$\{p[x := \textbf{new}]\} x := \textbf{new} \{p\},$$

where $x \in Var$ is a simple object variable and p is a pure assertion.

RULE 22: OBJECT CREATION

$$\frac{p' \rightarrow p[u := x]}{\{p'[x := \textbf{new}]\} u := \textbf{new} \{p\}}$$

where u is a subscripted normal object variable or a (possibly subscripted) instance object variable, $x \in Var$ is a fresh simple object variable which does not occur in p , and p' is a pure assertion.

RULE 23: SEQUENTIALIZATION

$$\frac{\{p\} S_1; \dots; S_n \{q\}}{\{p\} [S_1 \parallel \dots \parallel S_n] \{q\}}$$

RULE 24: DISJOINT PARALLELISM

$$\frac{\{p_i\} S_i \{q_i\}, i \in \{1, \dots, n\}}{\{\bigwedge_{i=1}^n p_i\} [S_1 \parallel \dots \parallel S_n] \{\bigwedge_{i=1}^n q_i\}}$$

where $free(p_i, q_i) \cap change(S_j) = \emptyset$ for $i \neq j$.

RULE 25: AUXILIARY VARIABLES

$$\frac{\{p\} S \{q\}}{\{p\} S_0 \{q\}}$$

where for some set of auxiliary variables A of S with $free(q) \cap A = \emptyset$, the program S_0 results from S by deleting all assignments to the variables in A .

RULE 26: ATOMIC REGION

$$\frac{\{p\} S \{q\}}{\{p\} \langle S \rangle \{q\}}$$

RULE 27: PARALLELISM WITH SHARED VARIABLES

$$\frac{\begin{array}{l} \text{The standard proof outlines } \{p_i\} S_i^* \{q_i\}, \\ i \in \{1, \dots, n\}, \text{ are interference free} \end{array}}{\{\bigwedge_{i=1}^n p_i\} [S_1 \parallel \dots \parallel S_n] \{\bigwedge_{i=1}^n q_i\}}$$

RULE 28: SYNCHRONIZATION

$$\frac{\{p \wedge B\} S \{q\}}{\{p\} \textbf{await } B \textbf{ then } S \textbf{ end } \{q\}}$$

RULE 29: PARALLELISM WITH DEADLOCK FREEDOM

$$\frac{\begin{array}{l} (1) \text{ The standard proof outlines } \{p_i\} S_i^* \{q_i\}, i \in \{1, \dots, n\} \\ \text{for weak total correctness are interference free,} \\ (2) \text{ For every potential deadlock } (R_1, \dots, R_n) \text{ of} \\ [S_1 \parallel \dots \parallel S_n] \text{ the corresponding tuple of} \\ \text{assertions } (r_1, \dots, r_n) \text{ satisfies } \neg \bigwedge_{i=1}^n r_i. \end{array}}{\{\bigwedge_{i=1}^n p_i\} [S_1 \parallel \dots \parallel S_n] \{\bigwedge_{i=1}^n q_i\}}$$

RULE 30: ALTERNATIVE COMMAND

$$\frac{\{p \wedge B_i\} S_i \{q\}, i \in \{1, \dots, n\}}{\{p\} \textbf{if } \Box_{i=1}^n B_i \rightarrow S_i \textbf{ fi } \{q\}}$$

RULE 31: REPETITIVE COMMAND

$$\frac{\{p \wedge B_i\} S_i \{p\}, i \in \{1, \dots, n\}}{\{p\} \textbf{do } \Box_{i=1}^n B_i \rightarrow S_i \textbf{ od } \{p \wedge \bigwedge_{i=1}^n \neg B_i\}}$$

RULE 32: ALTERNATIVE COMMAND II

$$\frac{\begin{array}{l} p \rightarrow \bigvee_{i=1}^n B_i, \\ \{p \wedge B_i\} S_i \{q\}, i \in \{1, \dots, n\} \end{array}}{\{p\} \textbf{if } \Box_{i=1}^n B_i \rightarrow S_i \textbf{ fi } \{q\}}$$

RULE 33: REPETITIVE COMMAND II

$$\frac{
\begin{array}{l}
\{p \wedge B_i\} S_i \{p\}, i \in \{1, \dots, n\}, \\
\{p \wedge B_i \wedge t = z\} S_i \{t < z\}, i \in \{1, \dots, n\}, \\
p \rightarrow t \geq 0
\end{array}
}{
\{p\} \text{ do } \bigwedge_{i=1}^n B_i \rightarrow S_i \text{ od } \{p \wedge \bigwedge_{i=1}^n \neg B_i\}
}$$

where t is an integer expression and z is an integer variable not occurring in p, t, B_i or S_i for $i \in \{1, \dots, n\}$.

RULE 34: DISTRIBUTED PROGRAMS

$$\frac{
\begin{array}{l}
\{p\} S_{1,0}; \dots; S_{n,0} \{I\}, \\
\{I \wedge B_{i,j} \wedge B_{k,\ell}\} \text{Eff}(\alpha_{i,j}, \alpha_{k,\ell}); S_{i,j}; S_{k,\ell} \{I\} \\
\text{for all } (i, j, k, \ell) \in \Gamma
\end{array}
}{
\{p\} S \{I \wedge \text{TERM}\}
}$$

RULE 35: DISTRIBUTED PROGRAMS II

$$\frac{
\begin{array}{l}
(1) \{p\} S_{1,0}; \dots; S_{n,0} \{I\}, \\
(2) \{I \wedge B_{i,j} \wedge B_{k,\ell}\} \text{Eff}(\alpha_{i,j}, \alpha_{k,\ell}); S_{i,j}; S_{k,\ell} \{I\} \\
\quad \text{for all } (i, j, k, \ell) \in \Gamma, \\
(3) \{I \wedge B_{i,j} \wedge B_{k,\ell} \wedge t = z\} \text{Eff}(\alpha_{i,j}, \alpha_{k,\ell}); S_{i,j}; S_{k,\ell} \{t < z\} \\
\quad \text{for all } (i, j, k, \ell) \in \Gamma, \\
(4) I \rightarrow t \geq 0
\end{array}
}{
\{p\} S \{I \wedge \text{TERM}\}
}$$

where t is an integer expression and z is an integer variable not appearing in p, t, I or S .

RULE 36: DISTRIBUTED PROGRAMS III

$$\frac{
\begin{array}{l}
(1) \{p\} S_{1,0}; \dots; S_{n,0} \{I\}, \\
(2) \{I \wedge B_{i,j} \wedge B_{k,\ell}\} \text{Eff}(\alpha_{i,j}, \alpha_{k,\ell}); S_{i,j}; S_{k,\ell} \{I\} \\
\quad \text{for all } (i, j, k, \ell) \in \Gamma, \\
(3) \{I \wedge B_{i,j} \wedge B_{k,\ell} \wedge t = z\} \text{Eff}(\alpha_{i,j}, \alpha_{k,\ell}); S_{i,j}; S_{k,\ell} \{t < z\} \\
\quad \text{for all } (i, j, k, \ell) \in \Gamma, \\
(4) I \rightarrow t \geq 0, \\
(5) I \wedge \text{BLOCK} \rightarrow \text{TERM}
\end{array}
}{
\{p\} S \{I \wedge \text{TERM}\}
}$$

where t is an integer expression and z is an integer variable not appearing in p, t, I or S .

AXIOM 37: RANDOM ASSIGNMENT

$$\{\forall x \geq 0 : p\} \ x := ? \ \{p\}$$

RULE 38: REPETITIVE COMMAND III

$$\frac{\begin{array}{l} \{p \wedge B_i\} \ S_i \ \{p\}, i \in \{1, \dots, n\}, \\ \{p \wedge B_i \wedge t = \alpha\} \ S_i \ \{t < \alpha\}, i \in \{1, \dots, n\}, \\ p \rightarrow t \in W \end{array}}{\{p\} \ \mathbf{do} \ \square_{i=1}^n B_i \rightarrow S_i \ \mathbf{od} \ \{p \wedge \bigwedge_{i=1}^n \neg B_i\}}$$

where

- (i) t is an expression which takes values in an irreflexive partial order $(P, <)$ that is well-founded on the subset $W \subseteq P$,
- (ii) α is a simple variable ranging over P that does not occur in p, t, B_i or S_i for $i \in \{1, \dots, n\}$.

RULE 39: FAIR REPETITION

$$\frac{\begin{array}{l} (1) \ \{p \wedge B_i\} \ S_i \ \{p\}, i \in \{1, \dots, n\}, \\ (2) \ \{p \wedge B_i \wedge \bar{z} \geq 0 \\ \quad \wedge \exists z_i \geq 0 : t[z_j := \mathbf{if} \ B_j \ \mathbf{then} \ z_j + 1 \ \mathbf{else} \ z_j \ \mathbf{fi}]_{j \neq i} = \alpha\} \\ \quad S_i \\ \quad \{t < \alpha\}, i \in \{1, \dots, n\}, \\ (3) \ p \wedge \bar{z} \geq 0 \rightarrow t \in W \end{array}}{(4) \ \{p\} \ \mathbf{do} \ \square_{i=1}^n B_i \rightarrow S_i \ \mathbf{od} \ \{p \wedge \bigwedge_{i=1}^n \neg B_i\}}$$

where

- (i) t is an expression which takes values in an irreflexive partial order $(P, <)$ that is well-founded on the subset $W \subseteq P$,
- (ii) $z_1, \dots, z_n$ are integer variables that do not occur in p, B_i or S_i , for $i \in \{1, \dots, n\}$,
- (iii) $t[z_j := \mathbf{if} \ B_j \ \mathbf{then} \ z_j + 1 \ \mathbf{else} \ z_j \ \mathbf{fi}]_{j \neq i}$ denotes the expression that results from t by substituting for every occurrence of z_j in t the conditional expression $\mathbf{if} \ B_j \ \mathbf{then} \ z_j + 1 \ \mathbf{else} \ z_j \ \mathbf{fi}$; here j ranges over the set $\{1, \dots, n\} - \{i\}$,
- (iv) $\bar{z} \geq 0$ abbreviates $z_1 \geq 0 \wedge \dots \wedge z_n \geq 0$,
- (v) α is a simple variable ranging over P and not occurring in p, t, B_i or S_i , for $i \in \{1, \dots, n\}$.

RULE 39': FAIR REPETITION (IDENTICAL GUARDS)

$$\begin{array}{l}
 (1') \{p \wedge B\} S_i \{p\}, i \in \{1, \dots, n\}, \\
 (2') \{p \wedge B \wedge \bar{z} \geq 0 \wedge \exists z_i \geq 0 : t[z_j := z_j + 1]_{j \neq i} = \alpha\} \\
 \quad S_i \\
 \quad \{t < \alpha\}, i \in \{1, \dots, n\}, \\
 (3') p \wedge \bar{z} \geq 0 \rightarrow t \in W \\
 \hline
 (4') \{p\} \mathbf{do} \Box_{i=1}^n B \rightarrow S_i \mathbf{od} \{p \wedge \neg B\}
 \end{array}$$

where conditions (i)–(v) of Rule 39 hold.

Auxiliary Axioms and Rules

RULE A1: DECOMPOSITION

$$\frac{\begin{array}{l} \vdash_p \{p\} S \{q\}, \\ \vdash_t \{p\} S \{\mathbf{true}\} \end{array}}{\{p\} S \{q\}}$$

where the provability signs $\vdash_p$ and $\vdash_t$ refer to proof systems for partial and total correctness for the considered program S , respectively.

AXIOM A2: INVARIANCE

$$\{p\} S \{p\}$$

where $free(p) \cap change(S) = \emptyset$.

RULE A3: DISJUNCTION

$$\frac{\{p\} S \{q\}, \{r\} S \{q\}}{\{p \vee r\} S \{q\}}$$

RULE A4: CONJUNCTION

$$\frac{\{p_1\} S \{q_1\}, \{p_2\} S \{q_2\}}{\{p_1 \wedge p_2\} S \{q_1 \wedge q_2\}}$$

RULE A5: $\exists$ -INTRODUCTION

$$\frac{\{p\} S \{q\}}{\{\exists x : p\} S \{q\}}$$

where x does not occur in S or in $free(q)$.

RULE A6: INVARIANCE

$$\frac{\{r\} S \{q\}}{\{p \wedge r\} S \{p \wedge q\}}$$

where $free(p) \cap change(S) = \emptyset$.

RULE A7: SUBSTITUTION

$$\frac{\{p\} S \{q\}}{\{p[\bar{z} := \bar{t}]\} S \{q[\bar{z} := \bar{t}]\}}$$

where $(\{\bar{z}\} \cap \text{var}(S)) \cup (\text{var}(\bar{t}) \cap \text{change}(S)) = \emptyset$.

RULE A8:

$$\frac{I_1 \text{ and } I_2 \text{ are global invariant relative to } p}{I_1 \wedge I_2 \text{ is a global invariant relative to } p}$$

RULE A9:

$$\frac{\begin{array}{l} I \text{ is a global invariant relative to } p, \\ \{p\} S \{q\} \end{array}}{\{p\} S \{I \wedge q\}}$$